Ajay Venkatesh

+1 516 585 9013 Email LinkedIn GitHub Portfolio

Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Sep 2024 – May 2026

- *Coursework:* System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Skills

Programming Languages: Python, Java, C++, TypeScript, JavaScript (ES6), SQL, C#
Web Technologies: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize, Django, FastAPI
Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs), Kubernetes
Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight
AI/ML: PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark

Experience

CampusMesh — **Jan 2026 – Present**

Software Engineer | *Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines*

- Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
- Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda, API Gateway, S3, and CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

Amazon — **May 2025 – Aug 2025**

Software Development Engineer Intern | *Serverless AWS infrastructure, React 18 frontend, and Java backend systems*

- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access **(4+ roles)** using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments.**
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; **applied low-level design and design patterns** reduced backend execution time by **35%** and ensure reliability by unit and integration testing using **Mockito**.

New York University (Novel AI Technologies, Inc.) — **Aug 2024 – Present**

Software Developer (On-Campus) | *NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics*

- Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrix analysis and process food images.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)**; deployed **(Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue.
- Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

PricewaterhouseCoopers (PwC) — **Aug 2021 – Aug 2024**

Senior Software Engineer | *Cloud-native backend, NLP/OCR automation, and data engineering on Azure*

- Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
- Deployed an **NLP** based **OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
- Optimized data ingestion with **parallel processing** for **100K**-row from Excel; reduced execution time from **5 hours to 1 hour**.
- Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

Personal Finance Analyzer (Python Data Analysis Project) | ***Python, Pandas, NumPy, Matplotlib, Seaborn***

- Built a data-driven application to analyze personal financial transactions and categorize expenses using **Pandas** for efficient data processing.
- Generated interactive visualizations and spending insights using **Matplotlib** and **Seaborn** to identify trends and optimize budgeting.
- Designed reusable and modular code structure following **OOP principles** to improve maintainability and scalability.

Certifications

AWS Cloud Practitioner	Azure Data Fundamentals	Microsoft Solution Architect Expert
Power BI Data Analyst Associate	Microsoft Developer Associate	Stanford Machine Learning (Coursera)